STAT 544 Study Guide and Lecture Prep: Conditional Distributions I

How to use this guide

Read Part 1 for the vocabulary, Part 2 for the derivation recipe, and Part 3 for conditional expectation. Then try each worked example in Part 4 yourself before reading the solution. Part 5 lists pitfalls and self-test questions. Part 6 gives R checks.

The one idea to hold onto: a conditional distribution is the joint distribution with one variable frozen, then rescaled so it integrates (or sums) to 1. Everything below is bookkeeping around that sentence.

Part 1. Definitions

1.1 Standard (undergraduate-level) definitions

Discrete case. For discrete X, Y with joint PMF $p_{X,Y}(x,y) = P(X=x, Y=y)$:

- Marginal: $p_X(x) = \sum_y p_{X,Y}(x,y)$.
- Conditional PMF of Y given $X=x$, defined only when $p_X(x) > 0$:
$$p_Y|X(y|x) = P(Y = y mid X = x) = frac p_X, Y(x,y) p_X(x).$$
- Validity: $p_{Y|X}(y|x) \geq 0$ and $\sum_y p_{Y|X}(y|x) = 1$ for each fixed x.

Continuous case. For jointly continuous X, Y with joint PDF $f_{X,Y}$:

- Marginal: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy$.
- Conditional PDF of Y given $X=x$, defined when $f_X(x) > 0$:
$$f_Y|X(y|x) = frac f_X, Y(x,y) f_X(x).$$
- Validity: $f_{Y|X}(y|x) \geq 0$ and $\int f_{Y|X}(y|x)\, dy = 1$.
- Probabilities: $P(Y \in B \mid X = x) = \int_B f_{Y|X}(y|x)\, dy$.

Product rule and independence.

$$f_X, Y(x,y) = f_Y|X(y|x), f_X(x) = f_X|Y(x|y), f_Y(y).$$

X and Y are independent iff $f_{Y|X}(y|x) = f_Y(y)$ for all x with $f_X(x) > 0$. The support must be a product set $A \times B$, so a triangular support always means dependence.

Bayes' rule for densities.

$$f_X|Y(x|y) = frac f_Y|X(y|x), f_X(x) int f_Y|X(y|x'), f_X(x'), dx'.$$

This is the engine of Bayesian inference later in the course. The denominator is just $f_Y(y)$.

Why the continuous definition is not "just conditional probability." Here $P(X = x) = 0$, so the elementary formula $P(A|B) = P(A \cap B)/P(B)$ is $0/0$. The density ratio is a definition chosen so that the product rule holds. The measure-theoretic definitions below show it is the right one, and also where it can go wrong.

1.2 Measure-theoretic definitions

Setting: a probability space $(\Omega, \mathcal{F}, P)$; random variables $X : \Omega \to \mathbb{R}^d$ and $Y : \Omega \to \mathbb{R}$ are measurable maps. The law of X is the pushforward $P_X(A) = P(X \in A)$.

Joint, marginal, and densities as Radon-Nikodym derivatives. The joint law $P_{X,Y}$ is a measure on $\mathbb{R}^{d+1}$. If $P_{X,Y} \ll \mu \otimes \nu$ for σ-finite μ, ν, then the density is

$$f_{X,Y} = \frac{dP_{X,Y}}{d(\mu \otimes \nu)}.$$

- Taking μ, ν as counting measure gives the PMF.
- Taking μ, ν as Lebesgue measure gives the PDF.
- Taking a mix gives mixed discrete and continuous pairs under one framework.

Fubini's theorem gives the marginal density $f_X(x) = \int f_{X,Y}(x, y)\, \nu(dy)$.

Conditional expectation given a sigma-algebra (Kolmogorov). Let $Y \in L^1(P)$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub-sigma-algebra. $E[Y|\mathcal{G}]$ is the (a.s. unique) random variable Z with

1. Z is $\mathcal{G}$-measurable, and
2. $\int_G Z\, dP = \int_G Y\, dP$ for every $G \in \mathcal{G}$.

Existence comes from Radon-Nikodym applied to the signed measure $G \mapsto \int_G Y\, dP$, which is absolutely continuous with respect to $P|_{\mathcal{G}}$. Uniqueness is only almost sure.

Conditioning on a random variable. $E[Y|X] := E[Y|\sigma(X)]$. By the Doob-Dynkin lemma there is a Borel function h with $E[Y|X] = h(X)$. We then define

$$E[Y|X = x] := h(x),$$

determined only P_X-almost everywhere. Different versions of h can differ on a P_X-null set of x values. For continuous X, the value at a single point x is therefore not canonically defined. The continuous formulas below give a good version.

Conditional probability is the special case $P(A|\mathcal{G}) = E[\mathbf{1}_A|\mathcal{G}]$.

Regular conditional distribution (probability kernel). A function $\kappa : \mathbb{R}^d \times \mathcal{B}(\mathbb{R}) \to [0, 1]$ such that

1. for each x, $\kappa(x, \cdot)$ is a probability measure;
2. for each Borel B, $x \mapsto \kappa(x, B)$ is measurable;
3. disintegration holds:
$$P(X \in A, , Y \in B) = \int_A \kappa(x, B)\, P_X(dx) \quad \text{for all Borel } A, B.$$

We write $\kappa(x, B) = P(Y \in B | X = x)$. A regular version exists whenever Y takes values in a Polish (Borel) space, so $\mathbb{R}$ and $\mathbb{R}^k$ are covered. For general Y, conditional probabilities exist for each B but may fail to assemble into a measure.

Link to densities. If $P_{X,Y} \ll \mu \otimes \nu$ with density $f_{X,Y}$, a valid kernel is

$$kappa(x, B) = int_B fracf_X, Y(x, y) f_X(x), nu(dy) qquad text for f_X(x) > 0.$$

Check: for any $g \geq 0$, $E[g(X)\mathbf{1}_B(Y)] = \iint g(x)\mathbf{1}_B(y) f_{X,Y}\, d\nu\, d\mu = \int g(x)\, \kappa(x, B)\, f_X(x)\, \mu(dx)$. This is exactly the disintegration identity, so the elementary definition is a theorem here.

Borel-Kolmogorov caution. Conditioning on a null event such as $\{X = Y\}$ depends on how the event is approximated. For example, conditioning on $Y - X = 0$ and on $Y/X = 1$ give different answers. The statement "condition on $X = x$" is meaningful only relative to the specific random variable X and the sigma-algebra $\sigma(X)$.

1.3 Translation table

Standard	Measure-theoretic
$p_{Y\|X}(y\|x)$ or $f_{Y\|X}(y\|x)$	density of $\kappa(x, \cdot)$ with respect to ν
$E[Y\|X = x]$	$h(x)$ where $E[Y\|\sigma(X)] = h(X)$, unique P_X-a.e.
"$f_X(x) > 0$"	x outside a P_X-null set
"integrating out"	Fubini on $\mu \otimes \nu$

Part 2. From joint to marginal to conditional: the recipe

Given a joint distribution, follow these steps every time.

1. **Draw the support.** Write the region where $f_{X,Y} > 0$ as inequalities. Note whether bounds of one variable depend on the other.
2. **Check normalization** (sum or integral equals 1). A cheap sanity check on the problem statement.
3. **Marginalize the conditioning variable.** For $f_X(x)$, integrate (or sum) over y, using y-limits that depend on x if the support is not a rectangle.
4. **State where $f_X(x) > 0$.** The conditional is defined only there.
5. **Divide:** $f_{Y|X}(y|x) = f_{X,Y}(x, y)/f_X(x)$. Keep the y-support, which may depend on x.
6. **Verify** that $\int f_{Y|X}(y|x)\, dy = 1$. If it is not 1, step 3 has an error.
7. **Compute what is asked:** a conditional probability (integrate over B), a conditional mean or variance (Part 3), or the other direction $f_{X|Y}$ by swapping roles.
8. **Cross-check** with the tower property $E[E[Y|X]] = E[Y]$, computed independently.

Shortcut: in step 5, the conditional density is proportional to the joint in y, with x held fixed. Anything depending only on x is part of the normalizing constant, so you can often identify a named distribution by inspecting the y-dependent kernel and skip the explicit marginal.

Part 3. Conditional expectation $E[Y|X=x]$

3.1 Formulas

$$E[Y|X=x] = \sum_y y\, p_{Y|X}(y|x) \qquad \text{or} \qquad \int y\, f_{Y|X}(y|x)\, dy.$$

For a function g: $E[g(Y)|X=x] = \int g(y) f_{Y|X}(y|x)\, dy$.

$E[Y|X=x]$ is a number for each x. $E[Y|X] = h(X)$ is a random variable. Confusing these is the most common mistake.

3.2 Properties (all hold almost surely)

1. **Linearity:** $E[aY_1 + bY_2|X] = aE[Y_1|X] + bE[Y_2|X]$.
2. **Taking out what is known:** $E[g(X)Y|X] = g(X)E[Y|X]$.
3. **Tower / law of total expectation:** $E[E[Y|X]] = E[Y]$. More generally, for $\mathcal{H} \subseteq \mathcal{G}$, $E[E[Y|\mathcal{G}]|\mathcal{H}] = E[Y|\mathcal{H}]$.
4. **Independence:** if $Y \perp X$, then $E[Y|X] = E[Y]$. The converse is false: zero dependence in the mean does not imply independence.
5. **Law of total variance:**
   $operatornameVar(Y) = E[operatornameVar(Y|X)] + operatornameVar(E[Y|X])$.
6. **Conditional variance:** $\text{Var}(Y|X=x) = E[Y^2|X=x] - (E[Y|X=x])^2$.
7. **Jensen:** $\varphi(E[Y|X]) \le E[\varphi(Y)|X]$ for convex φ.

3.3 Why it matters: the best predictor

If $E[Y^2] < \infty$, then $E[Y|X]$ is the orthogonal projection of Y onto the closed subspace of square-integrable functions of X. It uniquely minimizes

$$Ebig[(Y - g(X))^2 big] quad text over all measurable g.$$

This is the theoretical target of regression: the regression function is $E[Y|X=x]$. Linear regression is the special case where this function happens to be linear, as with the bivariate normal below.

3.4 Reference result: bivariate normal

If $(X, Y) \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$, then

$$Y|X=x sim N!Big(mu_Y + rho frac sigma_Y sigma_X (x - mu_X), sigma_Y^2(1 - rho^2) Big).$$

The conditional mean is linear in x, and the conditional variance is constant. This is a special feature of the normal.

Part 4. Four worked examples

Example 1 (discrete): a joint PMF table

Let $X \in \{1, 2, 3\}$ and $Y \in \{0, 1\}$ with joint PMF

	$x = 1$	$x = 2$	$x = 3$
$y = 0$	0.10	0.20	0.10
$y = 1$	0.20	0.15	0.25
Find the marginals, (p_{Y	X}), (E[Y	X=x]), verify the tower property, and find (E[X	Y=1]).

Step 1: normalization. $0.10 + 0.20 + 0.10 + 0.20 + 0.15 + 0.25 = 1.00$. Valid.

Step 2: marginals. Sum columns for X:
$p_X(1) = 0.10 + 0.20 = 0.30$, $p_X(2) = 0.20 + 0.15 = 0.35$, $p_X(3) = 0.10 + 0.25 = 0.35$.
Sum rows for Y: $p_Y(0) = 0.40$, $p_Y(1) = 0.60$.

Step 3: conditional PMF of Y given $X = x$. Divide each column by its total:

	$x = 1$	$x = 2$	$x = 3$
$P(Y = 0 \| X = x)$	$0.10/0.30 = 1/3$	$0.20/0.35 = 4/7$	$0.10/0.35 = 2/7$
$P(Y = 1 \| X = x)$	$0.20/0.30 = 2/3$	$0.15/0.35 = 3/7$	$0.25/0.35 = 5/7$

Each column sums to 1.

Step 4: conditional mean. Since Y is 0/1, $E[Y|X = x] = P(Y = 1|X = x)$:
$h(1) = 2/3,\ h(2) = 3/7,\ h(3) = 5/7$.

Step 5: tower check.

$$E[h(X)] = 0.30 cdot t frac{2}{3} + 0.35 cdot t frac{3}{7} + 0.35 cdot t frac{5}{7} = 0.20 + 0.15 + 0.25 = 0.60,$$

which equals $E[Y] = p_Y(1) = 0.60$. Verified.

Step 6: the other direction. $p_{X|Y}(x|1) = p(x, 1)/0.60$: $1/3,\ 1/4,\ 5/12$ (sum = 1). Then

$$E[X|Y = 1] = 1 cdot t frac{1}{3} + 2 cdot t frac{1}{4} + 3 cdot t frac{5}{12} = t frac{4 + 6 + 15}{12} = t frac{25}{12}$$

Takeaway. Conditioning on a column means renormalizing that column. X and Y are dependent because the conditional probabilities differ across x, and they differ from $P(Y = 1) = 0.6$.

Example 2 (continuous): $f(x,y) = x + y$ on the unit square

Let $f_{X,Y}(x,y) = x + y$ for $0 < x < 1,\ 0 < y < 1$. Find f_X, $f_{Y|X}$, $E[Y|X = x]$, and $P(Y < \tfrac{1}{2} \mid X = \tfrac{1}{2})$.

Step 1: normalization. $\int_0^1 \int_0^1 (x + y)\, dy\, dx = \int_0^1 (x + \tfrac{1}{2})\, dx = \tfrac{1}{2} + \tfrac{1}{2} = 1$. Valid. The support is a rectangle, but the density does not factor, so X, Y are dependent.

Step 2: marginal of X.

It is positive on the whole interval, so the conditional is defined for every $x \in (0,1)$.

Step 3: conditional density.

Check: $\int_0^1 \frac{x+y}{x+1/2} dy = \frac{x+1/2}{x+1/2} = 1$.

Step 4: conditional mean.

$E[Y|X = x] = frac1x + tfrac12int_0^1 y(x + y), dy = fracx/2 + 1/3x + 1/2 = frac3x + 26x + 3.$

Sanity values: $x \to 0$ gives $2/3$, and $x \to 1$ gives $5/9$. The mean of Y falls as x rises, which seems odd, since the joint density $x + y$ increases in both. It is correct: a larger x makes the x term dominate and flattens the y-shape toward uniform, whose mean is $1/2$.

Step 5: tower check.

$E[Y] = int_0^1(x + tfrac12)cdotfracx/2 + 1/3x + 1/2, dx = int_0^1left(tfracx2 + tfrac13right)dx$

Direct: $f_Y(y) = y + \frac{1}{2}$ by symmetry, so $E[Y] = \int_0^1 y(y + \frac{1}{2})dy = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$. Matches.

Step 6: a conditional probability at $x = \frac{1}{2}$. Here $f_X(\frac{1}{2}) = 1$, so $f_{Y|X}(y|\frac{1}{2}) = \frac{1}{2} + y$:

Example 3 (continuous, non-rectangular support): $f(x,y) = 8xy$ on $0 < x < y < 1$

Find f_X, f_Y, both conditional densities, both conditional means, and verify with $E[X]$.

Step 1: support and normalization. The region is the triangle above the diagonal. The bounds interact, so X, Y are dependent regardless of the density.

$$int_0^1int_0^y8xy, dx, dy = int_0^18ycdottfracy^22, dy = int_0^14y^3, dy = 1.$$

Valid.

Step 2: marginals. For fixed x, y runs from x to 1:

For fixed y, x runs from 0 to y:

Step 3: conditional densities. Keep the support limits.

Checks: $\int_0^y 2x/y^2\, dx = 1$. $\int_x^1 2y/(1 - x^2)\, dy = (1 - x^2)/(1 - x^2) = 1$.

Step 4: conditional means.

$$E[X|Y=y] = int_0^y x cdot frac{2x}{y^2}, dx = frac{2}{y^2} cdot frac{y^3}{3} = frac{2y}{3}.$$

$$E[Y|X=x] = int_x^1 y cdot frac{2y}{1-x^2}, dy = frac{2(1-x^3)}{3(1-x^2)} = frac{2(1+x+x^2)}{3(1+$$

The last step uses $1-x^3 = (1-x)(1+x+x^2)$ and $1-x^2 = (1-x)(1+x)$. Sanity: at $x \to 0$ it is $2/3$; at $x \to 1$ it is $2 \cdot 3/(3 \cdot 2) = 1$, which is correct, since y is squeezed to 1.

Step 5: tower check.

=\tfrac23E[Y]=\tfrac23\int_0^1 y\cdot4y^3,dy=\tfrac23\cdot\tfrac45=\tfrac8{15}.

]

Direct: $E[X] = \int_0^1 x \cdot 4x(1-x^2)\,dx = 4(\frac{1}{3} - \frac{1}{5}) = \frac{8}{15}$. Matches.

Takeaway. The conditional support depends on the conditioning value. Forgetting this and integrating over $(0,1)$ is the usual error.

Example 4 (hierarchical, discrete): Poisson thinning

Let $N \sim \text{Poisson}(\lambda)$, and given $N=n$, $X|N=n \sim \text{Binomial}(n,p)$. (Think of N arrivals, each kept independently with probability p.) Let $q = 1-p$. Find the marginal of X, the conditional distribution of N given $X=x$, and $E[N|X=x]$. Evaluate for $\lambda = 10$, $p = 0.3$, $x = 4$.

Step 1: joint PMF from the product rule. For $0 \le x \le n$:

$$p(n,x) = P(N=n), P(X=x|N=n) = frac{e^{-lambda} lambda^n}{n!} binom{n}{x} p^x q^{n-x} = frac{e^{-lambd}$$

Step 2: marginal of X. Sum over $n \ge x$ and let $m = n - x$:

So $X \sim \text{Poisson}(\lambda p)$, using $-\lambda + \lambda q = -\lambda p$.

Step 3: conditional of N given $X=x$. Divide:

$$p(n|x) = frac{e^{-lambda} lambda^n p^x q^{n-x}}{(x!(n-x)!) e^{-lambda p} (lambda p)^x / x!} = frac{e^{-lambda q} (lam$$

(using $\lambda^n p^x = \lambda^{n-x}(\lambda p)^x$ and $e^{-\lambda + \lambda p} = e^{-\lambda q}$). This says $N - X \mid X = x \sim \text{Poisson}(\lambda q)$, regardless of x. The number of discarded arrivals is independent of the number kept.

Step 4: conditional expectation.

$$E[N|X=x] = x + E[N-X|X=x] = x + lambda(1-p).$$

This is linear in x.

Step 5: numbers. With $\lambda = 10, p = 0.3, x = 4$: $E[N|X=4] = 4 + 10(0.7) = 11$. Tower check: $E[E[N|X]] = E[X] + \lambda q = \lambda p + \lambda q = \lambda = 10 = E[N]$. Verified.

Takeaway. Recognizing the kernel in n lets you skip normalization: after dividing, the only terms left in n are $(\lambda q)^{n-x}/(n-x)!$, which is a shifted Poisson kernel.

Part 5. Pitfalls and self-test

Common mistakes

1. Treating $E[Y|X=x]$ (a number) and $E[Y|X]$ (a random variable) as the same object.
2. Dropping the dependence of the conditional support on the conditioning value (Example 3).
3. Dividing by the wrong marginal: $f_{Y|X}$ divides by f_X, not f_Y.
4. Treating $f_{Y|X}(y|x)$ as a probability. It is a density in y, and can exceed 1.
5. Claiming independence from $E[Y|X] = E[Y]$ alone. That shows only mean independence.
6. Conditioning on null events such as $\{X = Y\}$ without specifying the limiting random variable (Borel-Kolmogorov).
7. Using the factorization test with a non-rectangular support.

Self-test (answers are short; derive them yourself first)

1. For $f(x,y) = 2$ on $0 < x < y < 1$, find $f_{Y|X}$ and $E[Y|X=x]$. Answer: $Y|X=x \sim \text{Unif}(x,1)$, mean $(1+x)/2$.
2. Show from the definition in 1.2 that $E[E[Y|\mathcal{G}]] = E[Y]$. Hint: take $G = \Omega$.
3. For the bivariate normal, derive $E[Y|X=x]$ from $f_{X,Y}/f_X$ by completing the square.
4. Use total variance to compute $\text{Var}(N)$ in Example 4 from $N \sim \text{Poisson}(\lambda)$ and verify against the conditional structure of X.
5. Explain in one paragraph why a regular conditional distribution exists for real-valued Y, and what could fail otherwise.

Lecture prep questions to bring

- Which version of $E[Y|X=x]$ does the course adopt when x is a single point of a continuous distribution?
- How does the Bayes' rule form here connect to posterior densities in the next unit?
- How is conditioning handled when X is mixed discrete and continuous?

Part 6. R reference checks

```
# Example 1: discrete table
P <- matrix(c(.10,.20,.10,
              .20,.15,.25), nrow = 2, byrow = TRUE,
            dimnames = list(y = 0:1, x = 1:3))
px <- colSums(P)                 # marginal of X
cond <- sweep(P, 2, px, "/")     # P(Y=y | X=x), columns sum to 1
cond["1", ]                      # E[Y|X=x] = P(Y=1|X=x)
sum(px * cond["1", ])            # tower: should be 0.60

# Example 2: continuous, numerical conditional mean
```

```
f  <- function(x, y) x + y
fx <- function(x) integrate(function(y) f(x, y), 0, 1)$value
h  <- function(x) integrate(function(y) y * f(x, y), 0, 1)$value / fx(x)
c(h(0.5), (3*0.5 + 2) / (6*0.5 + 3))       # both 7/12
integrate(Vectorize(function(x) fx(x) * h(x)), 0, 1)$value   # 7/12

# Example 3: triangular support
hx <- function(x) integrate(function(y) y * 2*y/(1 - x^2), x, 1)$value
c(hx(0.4), 2*(1 + .4 + .4^2) / (3*(1 + .4)))   # should agree

# Example 4: simulation check of thinning
set.seed(544)
lambda <- 10; p <- 0.3
N <- rpois(2e6, lambda); X <- rbinom(2e6, N, p)
mean(N[X == 4])                  # approx 11 = 4 + lambda*(1-p)
mean(X); var(X)                  # both approx lambda*p = 3
```

One-page summary

- Conditional = joint with one variable fixed, rescaled by the marginal: $f_{Y|X} = f_{X,Y}/f_X$.
- Measure-theoretically, it is a probability kernel $\kappa(x, \cdot)$ satisfying $P(X \in A, Y \in B) = \int_A \kappa(x, B) P_X(dx)$, unique P_X-a.e.
- $E[Y|X]$ is the $\sigma(X)$-measurable variable matching integrals of Y over every set in $\sigma(X)$, and it is the L^2-best predictor of Y from X.
- Recipe: support, normalize, marginalize, divide, verify the conditional integrates to 1, compute, cross-check with the tower property.
- Always carry the conditional support, and always run the tower check.